A 0.0067-mm^2 12-bit 20-MS/s SAR ADC Using Digital Place-and-Route Tools in 40-nm CMOS

Yao-Hung Tsai and Shen-Iuan Liu, *Fellow, IEEE*

***Abstract*— A 12-bit 20-MS/s asynchronous successive approximation register (SAR) analog-to-digital converter (ADC) is presented by using the digital place-and-route (DPR) tools. The macrocells for the capacitive digital-to-analog converter, the bootstrapped switch, and the dynamic comparator are presented. The custom standard cells for the dynamic SAR logic are also presented. By using the macrocells and the custom standard ones, the layout of this SAR ADC is completed by using the DPR tools. Several techniques are presented to improve the parasitic capacitances, the current density of the metal interconnections, and the nonideal effects caused by the DPR tools. This SAR ADC is fabricated in 40-nm CMOS technology and its active area is 0.0067 mm^2. To compare with the full-custom method, the proposed DPR flow has speeded up by a factor of 288 to complete the interconnection wires. Its power dissipation is 363 μW at 20 MS/s and the calculated Walden FoM is 23 fJ/c. step at Nyquist frequency.**

***Index Terms*— Analog-to-digital converter (ADC), bootstrapped switch, capacitive digital-to-analog converter, digital place-and-route (DPR), dynamic logic, successive approximation register (SAR).**

I. Introduction

THE Internet-of-Things (IoT) applications are rapidly growing, such as various sensor read-out circuits and sensor networks. For the IoT applications, the successive-approximation-register (SAR) analog-to-digital converters (ADCs) [1]–[14] are widely adopted because of low power and small area. For the advanced CMOS processes, since the design rules of the layout become more and more complex, it is difficult to generate the layout of the mixed-signal circuit in a short time. Furthermore, the layout should be redesigned if the layout of the ADC is migrated to other processes.

To realize the layout of the synthesizable SAR ADCs, the custom layout compilers [15], [16] and the commercial digital place-and-route (DPR) tools [17]–[21] have been presented. The custom layout compiler generates layout by considering the analog nonlinearities, such as offset, linearity, noise, and reliability. However, the analog designers have to define the routing instructions for the ADC. Therefore, it takes a long time to deal with placement and routing of the transistors and the interconnection wires. For instance, 400 h are required in [16]. By using the DPR tools, although it saves the place-and-route time, the nonlinearities may be induced to impact the sensitive analog circuits.

When the DPR tools are utilized, the following issues should be considered. First, the capacitor mismatch in a capacitive digital-to-analog converter (CDAC) will limit the accuracy of the ADC. When the process, the floorplan, the netlist, and the vendor tools are altered, the interconnections will vary and induce the parasitic capacitances. For example, once the driving circuits for the CDAC are revised, the interconnection wires between them will be rerouted and it will induce the unpredicted parasitic capacitances due to the DPR tools. Second, the bootstrapped switch [18] requires a power gating library, which may not be supported by every standard cell library. Third, the dynamic SAR logic in [22] uses the long-channel transistors to mitigate the leakage issue of the dynamic circuits. Fourth, the DPR tools use the metals with a narrow width to connect the cells. It may induce the limited current density and the IR drop. These issues will be discussed and the corresponding solutions will be given.

This article is organized as follows. Section II gives the circuit description of the proposed SAR ADC by using the DPR tools. Section III gives the experimental results. The conclusion is given in Section IV.

II. Circuit Description

Fig. 1(a) shows a 12-bit SAR ADC that is composed of the sampling switches, a CDAC, a comparator, the SAR logic, and the CDAC calibration circuit. The asynchronous SAR ADC uses the 1-bit redundancy and utilizes 20% of the sampling period for data acquisition. Its timing diagram of the clocks is shown in Fig. 1(b). The lowest four bits of CDAC are realized by single-ended capacitors and the other bits of CDAC are realized by differential ones to minimize the dynamic range loss [23].

A. Capacitive Digital-to-Analog Converter (CDAC)

The CDAC is realized by using both the unit-length and the unit-element capacitors [23]. For example, Fig. 2 shows a 4-bit CDAC where the last two-bit capacitors are realized by the length difference (i.e., Δ and 2Δ) between two metal strips and the first two-bit ones use the unit elements with 4Δ. Note that the long metal strips are avoided by paralleling the unit-element capacitors instead of the unit-length ones.

Manuscript received January 10, 2022; revised April 13, 2022; accepted April 22, 2022. Date of publication May 5, 2022; date of current version June 29, 2022. This work was supported by the Intelligent & Sustainable Medical Electronics Research Fund of National Taiwan University and in part by the Ministry of Science and Technology, Taipei, Taiwan. *(Corresponding author: Shen-Iuan Liu.)*

The authors are with the Graduate Institute of Electronics Engineering, and the Department of Electrical Engineering, National Taiwan University, Taipei 10617, Taiwan (e-mail: lsi@ntu.edu.tw).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TVLSI.2022.3170325.

Digital Object Identifier 10.1109/TVLSI.2022.3170325

1063-8210 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.

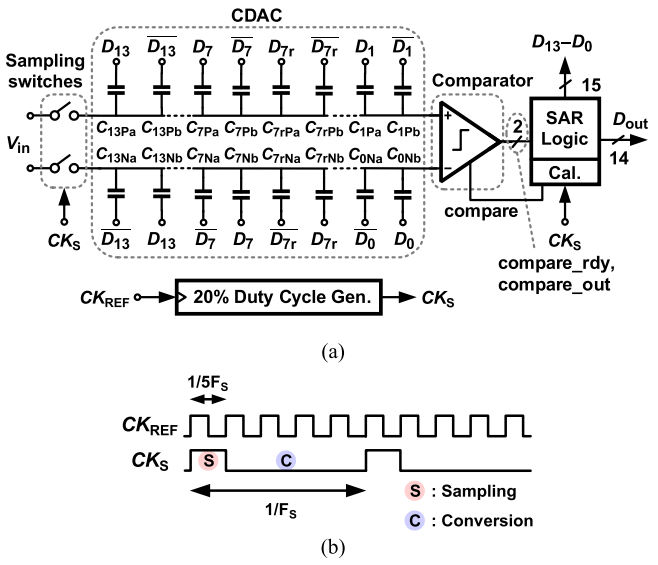


Fig. 1. (a) Twelve-bit SAR ADC and (b) its timing diagram of the clocks.

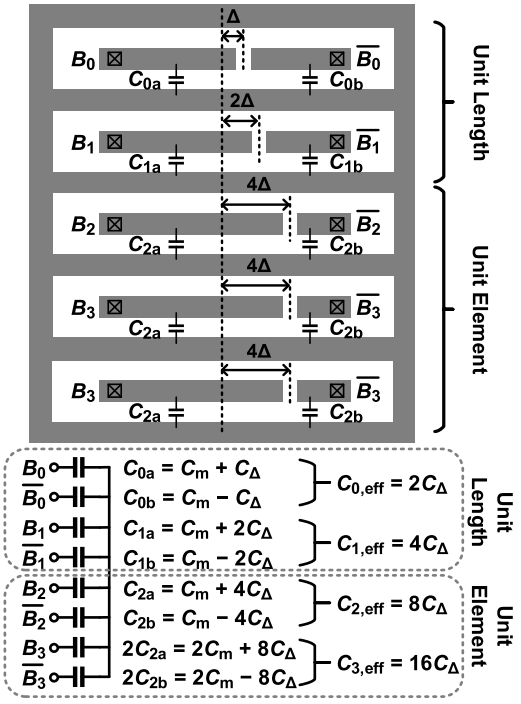


Fig. 2. Four-bit CDAC segmentation [7].

To explain the parasitic capacitances induced by the DPR tools, Fig. 3(a) shows a 3-bit CDAC. The main binary-weighted CDAC is realized by the 6th–8th metal layers. Since the CDAC is placed above the active circuits of the SAR ADC, the shield is built by interleaving and overlapping the fourth and fifth metal layers. The shield is used to prevent the crosstalk from the active circuits in the lower metal layers. By using the DPR tools, the metal interconnections are randomly routed between the capacitor and its control signal. For example, the relative position and the metal lengths are different for the control signals B_0–B_2 and $\overline{B_0}$–$\overline{B_2}$. The parasitic capacitances $C_{P,ka}$ and $C_{P,kb}$ are generated where $k = 0$–2. The effective capacitances controlled by B_0–B_2 are

TABLE I
SIMULATED 14-bit CDAC WITHOUT AND WITH EXTENDED SHIELD

Bit	Main CDAC	Parasitic capacitance	
		w/o extended shield	w/ extended shield
m	$2^{m+2}C_\Delta$ [fF]	$(C_{P,ma}-C_{P,mb})_{P,N}$ [fF]	$(C_{P,ma}'-C_{P,mb}')_{P,N}$ [fF]
0	0.11447	0.0011	−0.00119
1	0.22634	−0.0005	0.00044
2	0.45173	0.0004	−0.00056
3	0.90289	0.0008	−0.00014
4	1.80499	0.00004	0.00035
5	3.60864	0.0009	−0.00031
6	7.21633	0.1478	0.00035
7r	14.43336	−0.0204	0.000732
7	14.43315	−0.0117	0.000379
8	28.86708	−0.1086	0.000397
9	57.73243	0.134	0.00085
10	115.4654	−0.0135	0.00157
11	230.9264	−0.0486	0.01222
12	461.8275	0.0298	0.0429
13	923.6381	−0.1341	0.0791

expressed as

$$C_{k,\text{eff}} = 2^{k+1}C_\Delta + C_{P,ka} - C_{P,kb} \tag{1}$$

where C_Δ is the capacitance determined by Δ. Note that these asymmetrical parasitic capacitances will degrade the linearity and limit the accuracy of the SAR ADC. To reduce the undesired parasitic capacitances due to the DPR tools, Fig. 3(b) shows the proposed 3-bit CDAC with the extended shield. The extended shield is realized by the fourth and fifth metal layers to prevent the routing-induced parasitic capacitances between the fifth metal layer and the main CDAC.

In this work, the differential capacitance in the CDAC array is equal to $2^{m+2}C_\Delta$, where $m = 0$–13. This 14-bit CDAC array includes 12 bits for ADC conversion, two bits for CDAC calibration, and one redundant bit. The differential unit differential capacitor $4C_\Delta$ is equal to 114.47 aF. Table I shows the simulated parasitic capacitances of the 14-bit CDAC with and without the extended shield by using the DPR tools. Without the extended shield, the differences of the extracted parasitic capacitors for the sixth bit, the eighth bit, the ninth bit, and the most-significant bit (MSB) are 0.15, −0.11, 0.13, and −0.13 fF, respectively. With the extended shield, the differences of the extracted parasitic capacitors for the sixth bit, the eighth bit, the ninth bit, and the MSB are reduced to 0.00035, 0.000397, 0.00085, and 0.0791 fF, respectively. Fig. 4(a) and (b) shows the simulated differential nonlinearity (DNL) and integral nonlinearity (INL) of the CDAC array without and with extended shield, respectively. Without the extended shield, the peak DNL and INL are +0.38/−0.62 LSB and +0.73/−0.73 LSB, respectively. With the proposed extended shield, the peak DNL and INL are +0.01/−0.06 LSB and +0.04/−0.04 LSB, respectively. With the extended shield, the peak DNL and INL are reduced by 10 and 18 times, respectively.

Although the parasitic capacitances induced by the DPR tools are relaxed, the issues for the capacitor mismatch and the kT/C in the main CDAC must be considered. The CDAC

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.

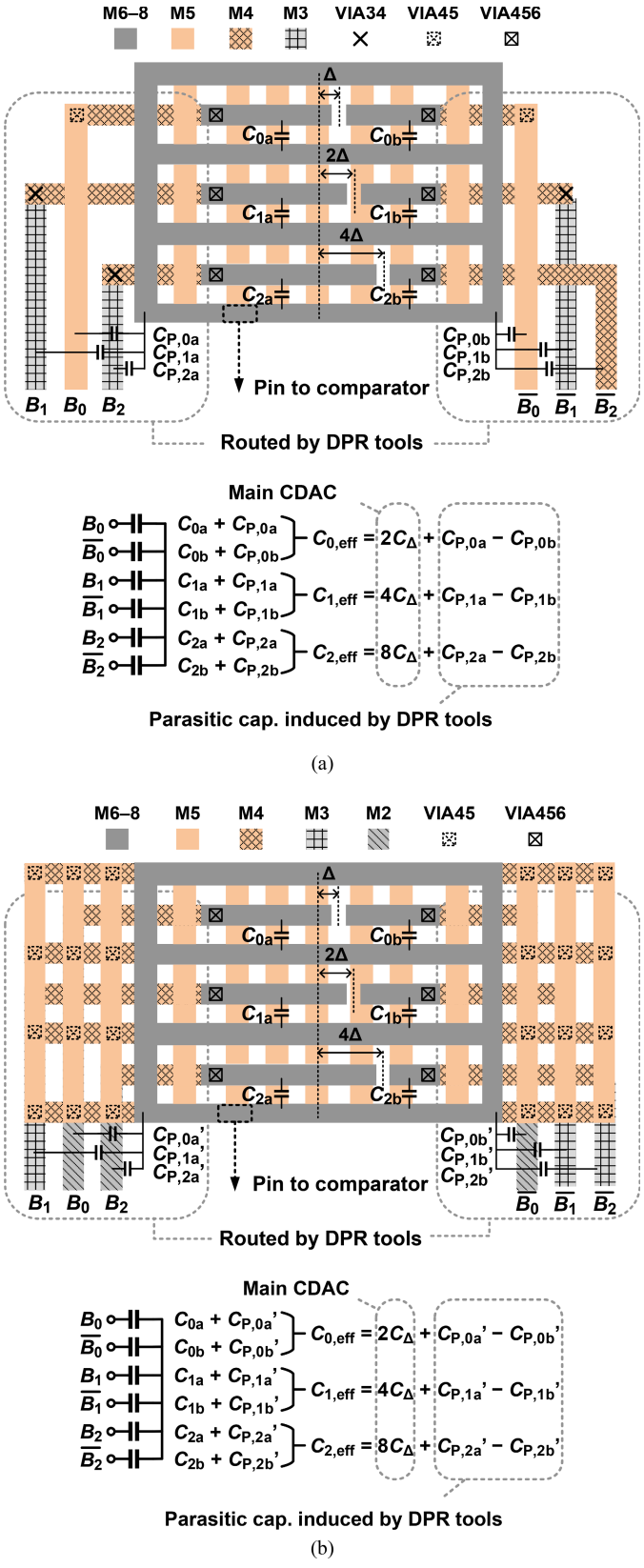


Fig. 3. Three-bit CDAC with (a) shield and (b) extended shield.

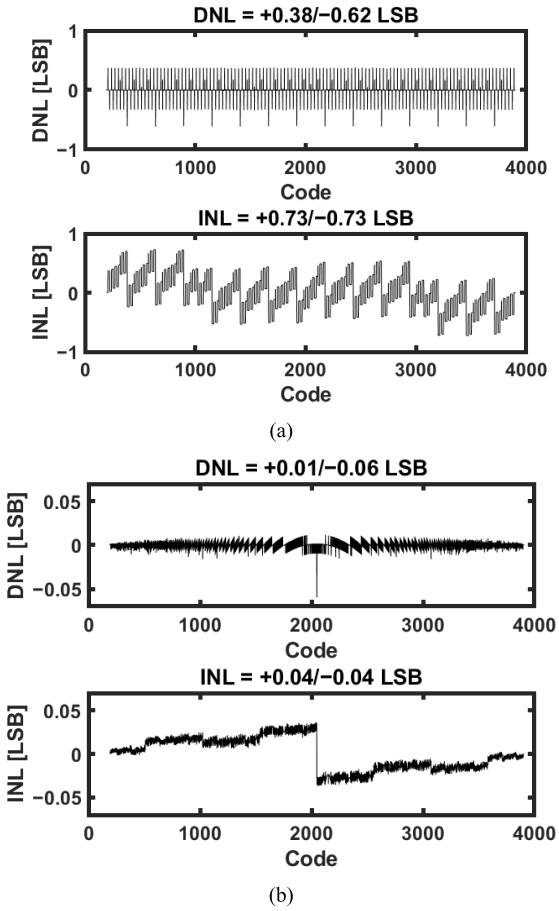


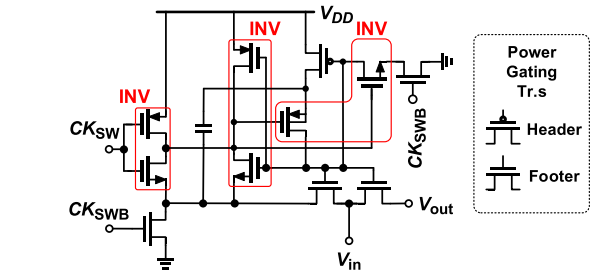
Fig. 4. Simulated DNL/INL of the CDAC array (a) without and (b) with extended shield.

area is designed for meeting the kT/C noise instead of the matching requirement. The single-sided capacitance of the CDAC in Fig. 1(a) is equal to 1.36 pF and the calculated kT/C noise voltage is 0.05 mV_{rms}. To improve the matching issue of the CDAC, the digital foreground CDAC calibration method [24] is adopted. Since the CDAC realized by the unit-length method is demonstrated in a 10-bit SAR ADC [23], the CDAC with eight least-significant bits (LSBs) acts as a reference one to calibrate the capacitor mismatches for the remaining 8–13 MSBs [24]–[25]. To realize the CDAC calibration algorithm, the circuits are synthesized by using the digital cell library.

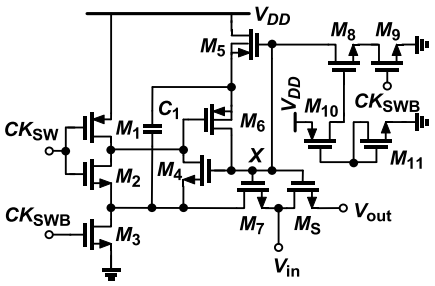
B. Bootstrapped Switch

Fig. 5(a) shows a synthesizable bootstrapped switch realized by the standard cells, including the power gating transistors and a decoupling capacitor [18]. To use the power gating transistors, a low-power DPR flow is required. The flow may not be supported by every standard cell library and require an additional tool license. Fig. 5(b) shows the proposed bootstrapped switch without the power gating transistors. It is composed of two custom standard cells and three macrocells. The standard cells can be placed and routed by the DPR tools; however, the macrocells cannot be placed automatically. The standard cells must meet some requirements such as cell height, bulk voltage definition, guard ring less, and power rail. For example, the width of the standard cells must be equal to the integral multiple of 0.14 μm in our work. The maximum height of the standard cells is limited to 1.26 μm by the

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.



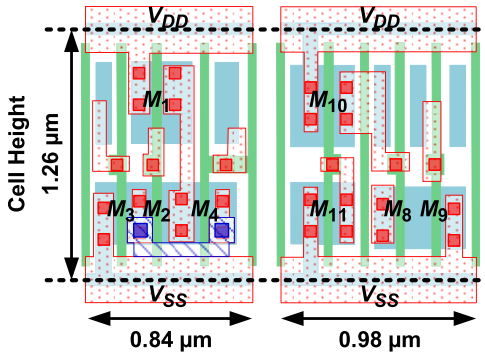
(a)



Device	Size
M_1, M_5–M_6, M_{10}	410 nm/ 40 nm
M_2–M_4, M_8–M_9, M_{11}	310 nm/ 40 nm
M_S	3.2 μm/ 40 nm
C_1	11.65 μm/ 30.17 μm

Device	Custom standard/macro cell	Dimension (Width × Height)
M_1–M_4	Custom standard (Dynamic NAND gate + Inverter)	0.84 × 1.26 μm²
M_8–M_{11}	Custom standard (Dynamic NAND gate + Tie-high)	0.98 × 1.26 μm²
M_5–M_6	Macro (Source-bulk tied)	2.1 × 1.68 μm²
M_7, M_S	Macro (Guard ring required)	2.72 × 2.58 μm²
C_1	Macro (Without power rail)	11.69 × 30.17 μm²

(b)



(c)

Fig. 5. (a) Synthesizable bootstrapped switch realized by standard cells including power gating transistors [4]. (b) Proposed switch realized by custom standard cells and macrocells. (c) Custom standard cells for M_1–M_4 and M_8–M_{11}.

foundry. Fig. 5(c) shows the layout of two custom standard cells. One is composed of M_1–M_4, which realizes a dynamic NAND gate and an inverter. The other custom standard cell is composed of a dynamic NAND gate M_8–M_9 and a tie-high cell M_{10} and M_{11}. Since the gate length of M_8 is short, there is a reliability issue if the gate of M_8 is directly connected to the supply voltage. The tie-high cell M_{10} and M_{11} is added to connect the gate of M_8 to the supply voltage through M_{10}.

To meet the analog circuit performance requirements, such as reliability, substrate noise, and area, the macrocells are used in this work. For the sake of reliability, the bulks of M_5 and M_6 are tied with their sources instead of the power supply. Then, a macrocell composed of M_5 and M_6 is generated. Another macrocell composed of M_S and M_7 are built with a guard ring to suppress the substrate noise. The metal–oxide–metal capacitor C_1 of 600 fF, which occupies 352.6873 μm^2, is realized by a macrocell without the power rails. To realize C_1 by using the conventional standard cell method, it occupies 609.6384 μm^2. By using the proposed macrocell, the area of C_1 is reduced by a factor of 1.7.

The bootstrapped switch [26] works in the track and hold modes. In the hold mode, when CK_{SW} = low and CK_{SWB} = high, the supply voltage V_{DD} is sampled on C_1. In the track mode, when CK_{SW} = high and CK_{SWB} = low, the voltage V_X at the node X is equal to $V_{in} + V_{DD}$, where V_{in} denotes the input voltage. Thus, the gate-to-source voltage of M_S, $V_{GS,MS}$, is theoretically constant [27]. However, when the frequencies of the input signal and the sampling clock are increased, the pole due to the ON-resistance of M_6, R_{ON6}, and the parasitic capacitor at the node X, C_X, will limit the swing. For an input signal $A_m \sin(\omega_{in} t)$, the output swing at the node X is approximated as

$$|V_X| = \frac{A_m}{\sqrt{1 + (R_{ON6} C_X \omega_{in})^2}} \tag{2}$$

where A_m is the amplitude and ω_{in} denotes the input frequency. When ω_{in} is $2\pi \times 10$ MHz and C_X is 8 fF, to meet the 12-bit linearity, the final normalized error of $|V_X|$ is 8×10^{-5}, R_{ON6} should be less than 25 kΩ, and R_{ON6} is decided to be 5.5 kΩ. The ON-resistance of M_7 can be determined in a similar way. The aspect ratios of M_6 and M_7 are chosen as 410 nm/40 nm and 310 nm/40 nm, respectively.

In the hold mode, the hold voltage $V_{C1,hold}$ of C_1 is determined as

$$V_{C1,hold} = V_{DD} \times \left(1 - e^{-\frac{t_{hold}}{R_{CG} C_1}}\right) \tag{3}$$

where R_{CG} is equal to sum of the ON-resistances of M_3 and M_5. When t_{hold} is 40 ns, C_1 is 600 fF, V_{DD} is 0.9 V, let $V_{C1,hold}$ settle within 8 ns, and R_{CG} should be less than 13.3 kΩ according to (3). The aspect ratios of M_3 and M_5 are 310 nm/40 nm and 410 nm/40 nm, respectively. To determine the aspect ratio of M_S, the similar effect of (2) is considered. To meet the 12-bit linearity, the ON-resistance of M_S is chosen as 130 Ω. The aspect ratio of M_S is 3.2 μm/40 nm. Fig. 6 shows the simulated signal-to-noise and distortion ratio (SNDR) and spurious-free dynamic range (SFDR) of the sample-and-hold circuit using the proposed bootstrapped switch with 2048-point FFT under various process corners at 0 °C and 120 °C, respectively. With an input frequency of 10 MHz and the sampling rate of 20 MS/s, the worst case simulated SNDR and SFDR under the slow–fast corner are 79.6 and 79.8 dB, respectively. Fig. 7 shows the simulated power spectrum of the sample-and-hold circuit using the proposed bootstrapped switch with 2048-point FFT under the slow–fast corner at 120 °C. The simulated SFDR is 79.8 dB.

Based on the above considerations and the simulation results, three macrocells and two custom standard cells are used to realize the proposed bootstrapped switch. The above

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.

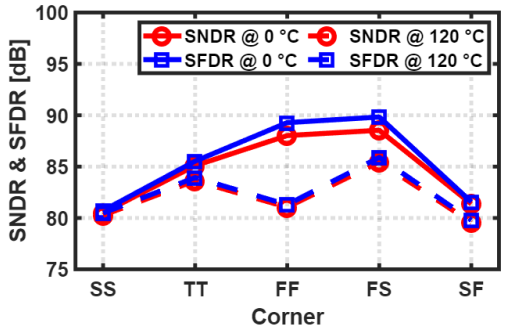


Fig. 6. Simulated SNDR and SFDR of the sample-and-hold circuit using the proposed bootstrapped switch with 2048-point FFT at $F_S = 20$ MS/s under various process corners at 0 °C and 120 °C.

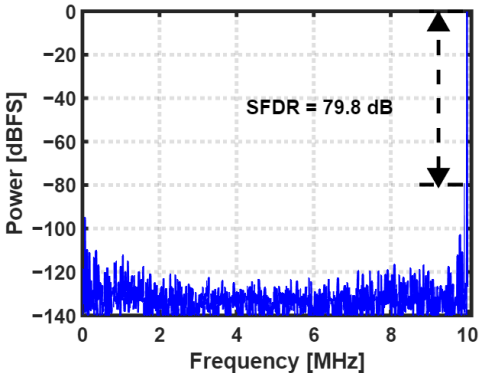


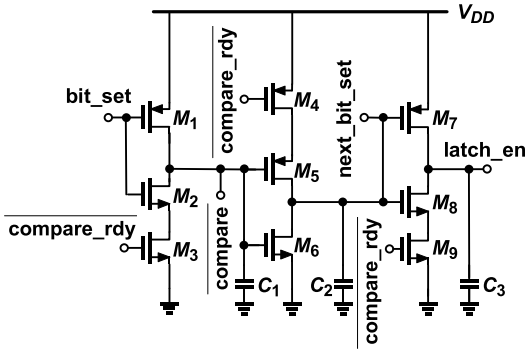
Fig. 7. Simulated power spectrum of the sample-and-hold circuit using the proposed bootstrapped switch with 2048-point FFT under the slow–fast corner at $F_S = 20$ MS/s and 120 °C.

cells are exported and added to the description files of the standard cell library (e.g., library exchange format (LEF) by using the Abstract Generator tool [28]).

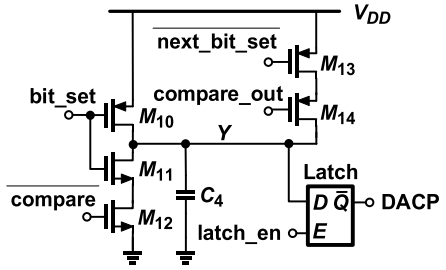
C. SAR Logic

The SAR logic may use the dynamic circuits to achieve low power dissipation. However, they are sensitive to the leakage current and the race condition. In this work, the SAR logic similar to [22] is adopted, except for the main control (MC) and DAC control (DC). The proposed MC and the DC are shown in Fig. 8(a) and (b), respectively. In [22], the MC uses the parasitic capacitors C_1–C_3 to store the digital control signal; however, C_1–C_3 are sensitive to the leakage current. When the signal bit_set is high, the proposed MC in Fig. 8(a) adopts the comparator ready signal, $\overline{\text{compare_rdy}}$, to periodically discharge C_1 for preventing M_6 from turning on due to the leakage current. Similarly, C_3 is also periodically discharged by $\overline{\text{compare_rdy}}$ to keep the latch enable signal, latch_en, low. C_2 is periodically charged by $\overline{\text{compare_rdy}}$ to prevent M_7 from turning on.

To avoid the race condition, the proposed DC is shown in Fig. 8(b). For example, if both bit_set and $\overline{\text{compare}}$ are high, Y goes low. When $\overline{\text{next_bit_set}}$ is low, compare_out goes low or high, the correct Y will be high or low, respectively. Once latch_en is disabled to low by the MC, the correct Y is latched as the signal DACP to control the CDAC array. Thus, the race issue in [22] is avoided. By using this latch, the parasitic capacitor C_4 is used instead of an additional MOS capacitor



(a)



(b)

Custom standard cell	Circuit
M_1–M_3, M_7–M_9 & M_{10}–M_{12}	N-type current-starved inverter
M_4–M_6	P-type current-starved inverter
M_{13}–M_{14}	Dynamic NOR gate

(c)

Fig. 8. Proposed (a) MC, (b) DC, and (c) custom standard cells.

here. Note that an MOS capacitor is required to avoid the leakage issue in [22].

To synthesize the SAR logic, three kinds of digital cells are realized, as shown in Fig. 8(c). An n-type current-starved inverter is realized as a custom standard cell to synthesize M_1–M_3, M_7–M_9, and M_{10}–M_{12}. A p-type current-starved inverter is used to realize M_4–M_6. The dynamic NOR cell is generated to realize M_{13}–M_{14}. Note that the SAR logic except MCs and DCs is synthesized by the standard cell library.

D. Comparator

Fig. 9 shows a double-tail dynamic comparator that is composed of a preamplifier and a latch. The loading capacitors at A_+ and A_- are not shown. To adopt the double-tail dynamic comparator as a macrocell instead of the standard cells [17]–[18], there are three kinds of considerations. First, the input common-mode voltage can be equal to half of the supply voltage to maximize the dynamic range of the SAR ADC. Second, the size of the preamplifier and the latch can be scaled for power efficiency and speed considerations. Third, to prevent the substrate noise, a guard ring is required. The layout of a standard cell does not adopt the guard ring.

By using the DPR tools to interconnect the dynamic comparator with other circuits, the automatically routed metals may induce the unbalanced parasitic capacitors at A_+/A_- and $V_{\text{out}+}/V_{\text{out}-}$ to degrade the input offset voltage [29] of

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.

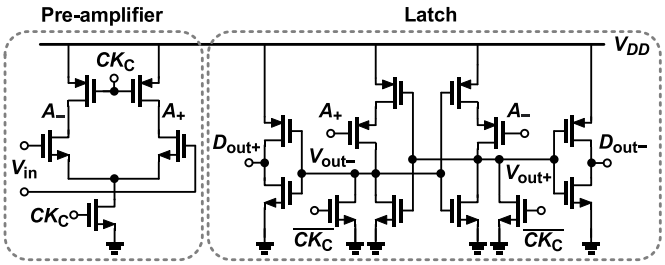


Fig. 9. Double-tail dynamic comparator.

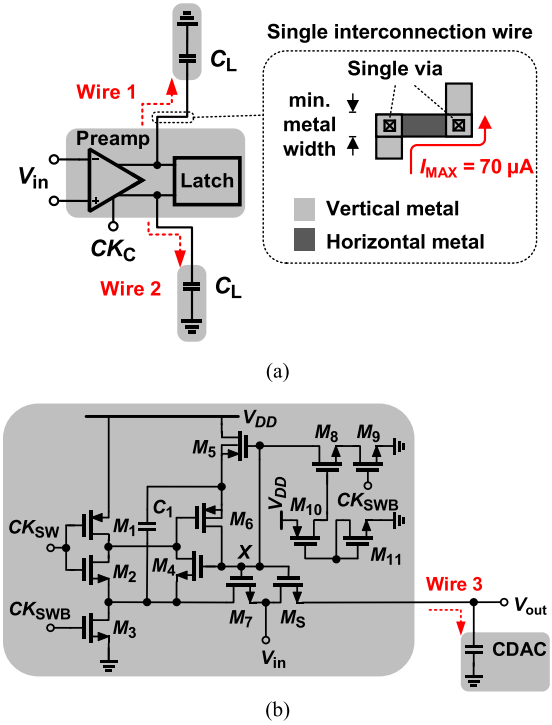


Fig. 10. Single interconnection wire of (a) comparator and (b) bootstrapped switch and the CDAC.

the dynamic comparator in Fig. 9. To relief this effect, the blockage constraint is added by using the Abstract Generator tool [28]. The coverage of the layout of the sensitive analog circuit is set where the DPR tools will not route the metal interconnections over this area. In this way, the parasitic capacitors caused by the DPR tools are reduced for the sensitive analog circuits. With the blockage constraint, the difference in the extracted parasitic capacitances between A_+ and A_- is equal to 0.24 fF, which is small enough to be ignored. The simulated input-referred offset voltage is equal to 0.05 mV for the dynamic comparator.

E. Current Density and IR Drop

By using the DPR tools to connect the analog/mixed-signal circuits, the resistance and the current density issues should be considered. In [21], the number of VIAs is increased to deal with the resistance issue. However, the current density of the metals and VIAs are not considered. Fig. 10(a) shows that there are two interconnection wires between the preamplifier with the latch and the loading capacitors C_L in the dynamic double-tail comparator. Note that the capacitor C_L is realized as a macrocell. By using the DPR tools, the vertical and

TABLE II
THREE CURRENT-DRIVEN WIRES

Wire	C	ΔV	$\bar{I}$	Δt
1, 2	200 fF	0.9 V	93.6 μA ($> I_{MAX}$)	$\frac{1}{20\text{ M}}:\frac{1}{2}:\frac{1}{13}$
			187.2 μA ($> I_{MAX}$)	$\frac{1}{40\text{ M}}:\frac{1}{2}:\frac{1}{13}$
3	1360 fF	0.9 V	48.96 μA	$\frac{1}{20\text{ M}}:\frac{1}{2}$
			97.92 μA ($> I_{MAX}$)	$\frac{1}{40\text{ M}}:\frac{1}{2}$

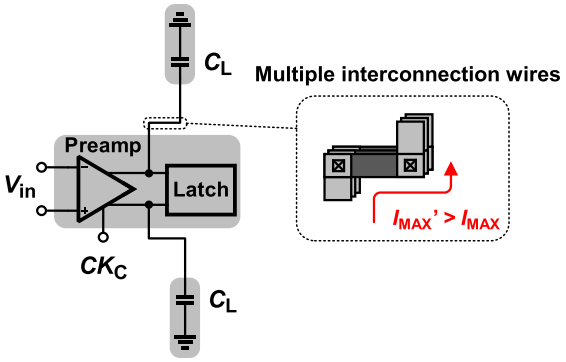


Fig. 11. Multiple interconnection wires for the comparator in the DPR flow.

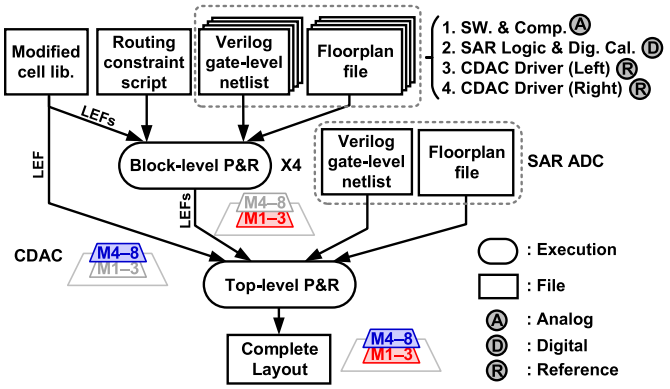
horizontal routings usually utilize the different metal layers. Besides, the metal with a minimum width and single VIA is used for the voltage-driven routing. For a 40-nm process, the maximum current I_{MAX} for the minimum width metal and single VIA is limited to 70 μA. Fig. 10(b) shows another interconnection wire in the proposed bootstrapped switch. During the time of Δt, the averaging current of the capacitor is approximated as

$$\bar{I} = \frac{C \times \Delta V}{\Delta t} \tag{4}$$

where C is the capacitance and ΔV is the voltage difference of the capacitor. The averaging currents passing the three interconnection wires are calculated and summarized in Table II. The capacitances C_L and the CDAC are equal to 200 fF and 1.36 pF, respectively. Δt is estimated by the reciprocal of the sampling frequency multiplied by the charging or discharging operation times. ΔV is equal to the supply voltage of 0.9 V. For every conversion cycle of the ADC, the comparator operates 13 times, including the 12 bits and one redundant bit. When the sampling rate is equal to 40 MS/s, the average current of the interconnection wires of Fig. 10(a) and (b) is equal to 187.2 and 97.92 μA, respectively. By using the DPR tools, both averaging currents exceed the maximum one of 70 μA. In Fig. 10(a), when the sampling rate is equal to 20 MS/s, the average current of the interconnection wire is equal to 93.6 μA, which also exceeds 70 μA.

To reduce the average current per interconnection wire, one can parallel the multiple interconnection wires between the preamplifier with the latch and the loading capacitor C_L, as shown in Fig. 11. The number of the multiple interconnection wires should be much higher than the ratio of the

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.



(a)

File	Setup time per file	# of file
Modified cell library	17 hours	1
Routing constraint script	1 minute	1
Verilog gate-level netlist	10 minutes	5
Floorplan file	20 minutes	5

(b)

Fig. 12. (a) Proposed DPR flow for the SAR ADC and (b) setup time in the DPR flow.

averaging current to I_{MAX}. In this work, four interconnection wires are used in the dynamic comparator. Similarly, the multiple interconnection wires are also considered for the proposed bootstrapped switch.

Assume that there is an IR drop of 90 mV in the power path and the simulated settling time of the comparator and the SAR logic is increased up to 34.275 ns, which is close to the operation period of 40 ns. The simulated average currents of the comparator and the SAR logic are 2 and 1 mA, respectively. For the comparator and the SAR logic with the IR drop of 90 mV, the resistances of their power paths should be less than 45 and 90 Ω, respectively. The equivalent lengths of the power rails of the comparator and SAR logic are about 63 and 88 μm, respectively. To consider the metal sheet resistance of 0.225 Ω/□, the metal widths of the power paths must be larger than 0.315 and 0.22 μm. In this work, the widths of the power rails are chosen as 0.46 μm for the comparator and SAR logic.

F. DPR Flow

In [18], by using the hierarchy DPR flow, the building blocks in the SAR ADC can be independently optimized for the sake of performances. However, the CDAC uses the 4th–8th metal layers and other circuits use the 1st–3rd metal layers in this work. Note that the CDAC is placed above the core circuits in order to save the area. The hierarchy DPR flow does not support it. Fig. 12(a) shows the proposed DPR flow. Before the place and route (P&R) executions, there are several files should be prepared, including a modified cell library, a routing constraint script, five Verilog gate-level netlists, and five floorplan files. First, the cell library is modified by adding six macrocells and five custom standard ones. The first macrocell is the CDAC with extended shield in Fig. 3(b). There are three macrocells and two custom standard ones for the bootstrapped switch in Fig. 5(b). To realize the SAR logic in Fig. 8, three custom standard cells are required. Two macrocells are used to realize the dynamic comparator in Fig. 10(a). Second, the routing constraint script is generated to constraint the P&R execution with the 1st–3rd metal layers. Third, three power domains, which are analog, digital, and reference, are adopted for four building blocks to isolate the digital supply noise. To minimize routing distances between CDAC and two CDAC drivers, two CDAC drivers are generated separately. Thus, four Verilog gate-level netlists and four floorplan files are required. The block-level P&R is executed four times. To combine the CDAC with all the building blocks, the last netlist and floorplan files for the SAR ADC are used. Finally, the top-level P&R is executed to generate the final layout of this SAR ADC. In the top-level P&R, all the metal layers are used.

The setup time for each file in the DPR flow is shown in Fig. 12(b). First, each cell is realized in three steps; drawing the layout, verifying design rule check (DRC)/layout versus schematic (LVS) by using the Calibre verification tool [30], and translating into the LEF files by using the Abstract Generator tool [28]. The setup time is equal to 3 and 10 h for the CDAC array and the macrocells of the comparator, respectively. For the remaining three macrocells and five custom standard cells, the setup time is equal to 30 min/cell. It totally takes 17 h for the modified cell library. Second, the routing constraint script is generated by the Encounter tool [31] with the setup time of 1 min. Third, each Verilog gate-level netlist requires the setup time of 10 min by using the Virtuoso NC-Verilog tool [32]. Finally, each floorplan file requires 20 min by using the Encounter tool [31]. The total setup time for the SAR ADC in the proposed DPR flow is equal to 19.5 h.

There are 62 interconnection wires between the CDAC array and its drivers. To realize the bootstrapped switches, 40 interconnection wires are required. The SAR logic uses 195 interconnection wires. There are totally 297 interconnection wires required for the proposed SAR ADC. By using the full-custom method, it averagely takes 5 min to complete one interconnection wire in the 40-nm CMOS process (including the time for DRC/LVS). To complete these interconnection wires, 1485 min (about 24 h) are required. Assume that the modified cell library, the routing constraint scripts, the Verilog gate-level netlists, and the floorplan files are ready. By using the DPR tools, the total runtime is within 5 min to complete the block-level P&R and the top-level one in our DPR flow. To compare with the full-custom method, the proposed DPR flow can speed up by a factor of 288 to complete the interconnection wires. To use the proposed DPR flow to generate the graphic design system II (GDSII) layout of the SAR ADC, it takes less than 20 h.

In summary, the potential issues are considered in the proposed DPR flow. First, the DPR tools may induce the parasitic capacitances, which degrades the linearity of the CDAC array and the offset of the comparator. Second, to maintain the performance of the analog circuits by using the DPR tools, the custom standard cells and macrocells are used for considering the reliability, the substrate noise, and the area. Third, the dynamic SAR logics are realized by the custom standard cells to mitigate the leakage issue. Finally, the limited current

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.

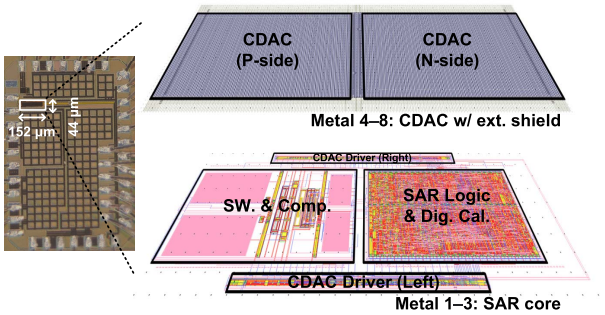


Fig. 13. Die photograph and layout of the SAR ADC.

density and the IR drop issues induced by using the narrow metal widths in the DPR tools have to be considered.

III. Experimental Results

The proposed SAR ADC is fabricated in the one-poly ten-metal 40-nm CMOS technology. Its die photograph and the magnified view of the layout are shown in Fig. 13. The ADC core occupies 44 μm $\times$ 152 μm. The measurement results are obtained with the supply voltage of 0.9 V and at room temperature. Fig. 14(a) shows the measured static performance without CDAC calibration at 20 MS/s. The measured DNL is within $+1.65/-0.98$ LSB and the measured INL is within $+2.25/-3.03$ LSB. Fig. 14(b) shows the measured static performance with CDAC calibration. The measured DNL is within $+2.08/-0.96$ LSB and the measured INL is within $+1.24/-1.41$ LSB.

There are several factors to degrade the DNL/INL of the SAR ADC, i.e., the capacitor mismatch in the CDAC array and the reference voltage ripple [33]. In this work, the INL is improved by using the CDAC calibration and the DNL is not. To understand the above problem, the following simulation results are given. Assume that the differential normalized capacitance ratios of the CDAC array from MSB to LSB are equal to $4090, 2049.5, 1025, 512.5, 256, 128, 64, 64, 32, \ldots,$ and 2. Without the CDAC calibration, the reference voltage is assumed to have a sinusoidal noise of 20 MHz with an amplitude of 1.8 mVpp. The simulated DNL is within $+1.54/-1$ LSB and the simulated INL is within $+1.64/-2.98$ LSB. When the CDAC calibration is activated, assume that the amplitude of the sinusoidal noise of 20 MHz is increased up to 2.6 mVpp. The simulated DNL is within $+2.07/-1$ LSB and the simulated INL is within $+1.11/-2.15$ LSB. Thus, the INL is improved by using the CDAC calibration. The DNL is degraded by the additional sinusoidal supply noise when the CDAC calibration is activated.

When the input frequency is 0.88 MHz and the sampling rate is 20 MS/s, Fig. 15(a) shows that the measured SNDR and SFDR are 59.1 and 64.2 dB, respectively, without CDAC calibration. Fig. 15(b) shows that the measured SNDR and SFDR are 63 and 75.1 dB, respectively, with CDAC calibration. When the input frequency is 9.73 MHz and the sampling rate is 20 MS/s, Fig. 16(a) shows that the measured SNDR and SFDR are 57.7 and 66.5 dB, respectively, without CDAC calibration. Fig. 16(b) shows that the measured SNDR and SFDR are 59.5 and 72.4 dB, respectively, with CDAC calibration.

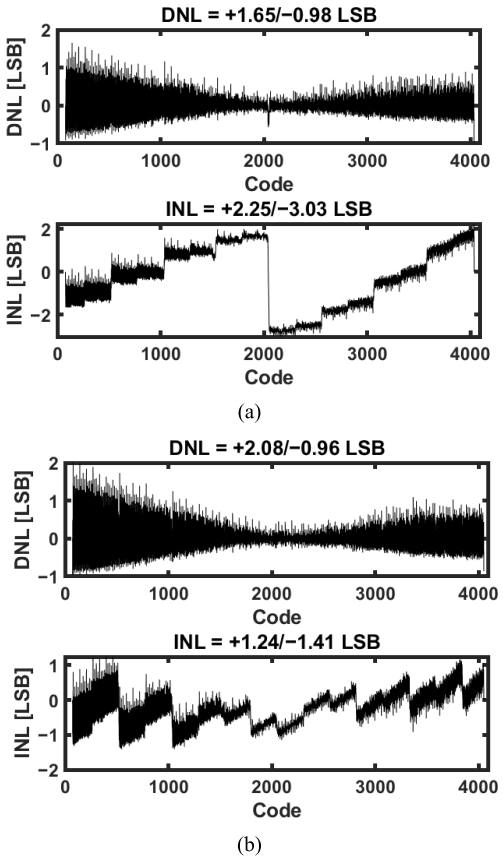


Fig. 14. Measured DNL and INL: (a) without and (b) with CDAC calibration.

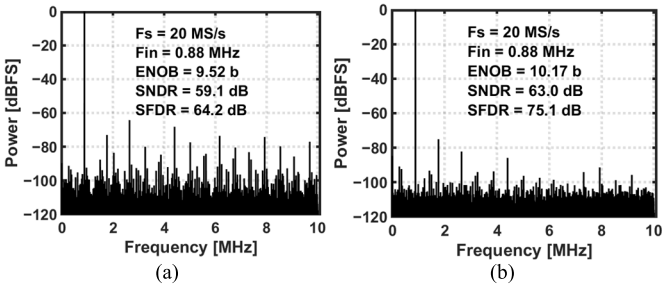


Fig. 15. FFT plot with 0.88-MHz input frequency at 20 MS/s: (a) without and (b) with CDAC calibration.

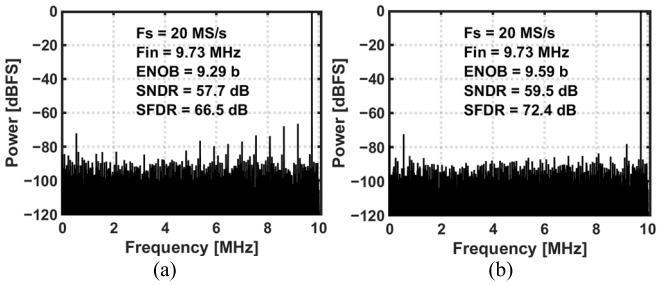


Fig. 16. FFT plot with 9.73-MHz input frequency at 20 MS/s: (a) without and (b) with CDAC calibration.

Fig. 17 plots the measured SNDR and SFDR versus the input frequency at 20 MS/s. At a low input frequency, the measured SNDR and SFDR are 63 and 75.1 dB, respectively, with CDAC calibration. When the input frequency goes to Nyquist, the

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PERFORMANCE AND COMPARISON SUMMARY

	[1]	[2]	[13]	[14]	[17]	[18]	This Work
Process [nm]	180	180	40	40	28	28	40
Area [mm^2]	0.359	1.61	0.0675	0.01	0.002	0.0086	0.0067
Supply [V]	1.8	1.8	1	0.9	0.5	1	0.9
Resolution	12 b	12 b	13 b	12 b	8 b	11 b	12 b
DNL [LSB]	+0.36/−0.33	+0.65/−0.71	+1.08/−	+0.94/−0.72	−	+0.23/−0.4	+2.08/−0.96
INL [LSB]	+0.55/−0.6	+0.98/−1.01	+3.79/−	+1.8/−1.8	−	+0.68/−0.68	+1.24/−1.41
SNDR @ Low freq. [dB]	70	65.2	65	62	34.2	58	63
SNDR @ Nyq. [dB]	66.9	64.6	64.1	64.1	28.5	56.8	59.5
SFDR @ Nyq. [dB]	75.8	75.5	81.9	75.6	−	69.2	72.4
Power [μW]	820	1770	46	340	92	399	363
Sample Rate [MS/s]	10	20	6.4	50	0.5	50	20
FoM_{Walden} @ Low freq. [fJ/c.step]	31.7	59.5	4.9	6.6	4391	12.3	15.7
FoM_{Walden} @ Nyq. [fJ/c.step]	45.3	63.7	5.5	5.3	8466	14.1	23.5
$FoM_{Schreier}$ @ Nyq. [dB]	164.8	162.1	172.4	172.8	128.5	164.7	163.9
DAC Calibration	No	No	Yes	No	Yes	No	Yes
Synthesizable	No	No	No	No	Yes	Yes	Yes

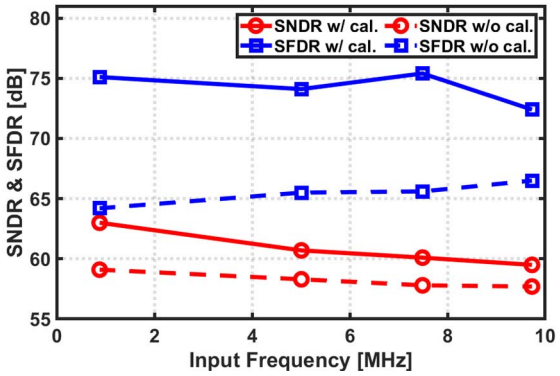


Fig. 17. Measured SNDR and SFDR versus input frequency at 20 MS/s.

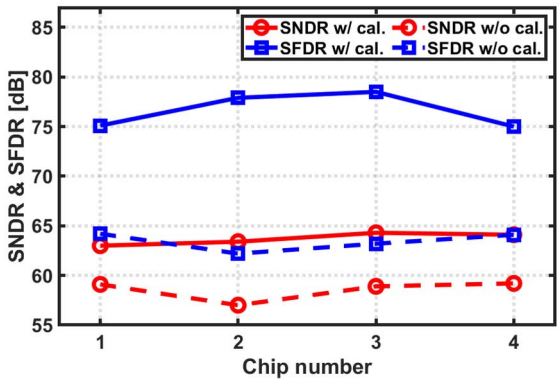


Fig. 18. Measured SNDR and SFDR for four chips.

measured SNDR and SFDR are 59.5 and 72.4 dB, respectively, with CDAC calibration. Fig. 18 shows the measured SNDR and SFDR for four chips at 0.88-MHz input and 20-MS/s sampling frequency. The measured SNDR and SFDR are averagely improved by 5.15 and 13.2 dB, respectively, with CDAC calibration. Table III compares the performance with several SAR ADCs [1], [2], [13], [14], [17], [18], with the similar speed and resolution. In [1] and [2], a supply voltage of 1.8 V is adopted for achieving the better SNDR and SFDR. However, their power consumption is increased. By using a supply voltage of 0.9 V, the proposed SAR ADC achieves a better FoM compared with [1] and [2]. In [13] and [14], good FoMs are achieved and the full-custom layouts are required. In [17], a fully synthesizable SAR ADC is presented; however, its FoM is poor. This work achieves the FoM similar to [18].

IV. CONCLUSION

The 12-bit 20-MS/s SAR ADC by using the DPR tools is presented. By using the proposed macrocells and the custom standard cells, the layout of this SAR ADC can be completed by using the DPR tools. To deal with the parasitic capacitance and the current density issues caused by the DPR tools, several techniques have been presented. It not only maintains the performance of the analog circuits but also saves the layout time. Finally, by using a 40-nm CMOS process, the experimental results of the SAR ADC are also demonstrated.

ACKNOWLEDGMENT

The authors would like to thank TSMC University-shuttle program to fabricate this chip.

REFERENCES

[1] S. Liu, Y. Shen, and Z. Zhu, "A 12-bit 10 MS/s SAR ADC with high linearity and energy-efficient switching," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 63, no. 10, pp. 1616–1627, Oct. 2016.

[2] Y. Shen, Z. Zhu, S. Liu, and Y. Yang, "A reconfigurable 10-to-12-b 80-to-20-MS/s bandwidth scalable SAR ADC," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 65, no. 1, pp. 51–60, Jan. 2018.

[3] Y. Liang, C. Li, S. Liu, and Z. Zhu, "A 14-b 20-MS/s 78.8dB-SNDR energy-efficient SAR ADC with background mismatch calibration and noise-reduction techniques for portable medical ultrasound systems," *IEEE Trans. Biomed. Circuits Syst.*, early access, Feb. 1, 2022, doi: 10.1109/TBCAS.2022.3147954.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.

[4] A. Akkaya, F. Celik, and Y. Leblebici, "An 8-bit 800 MS/s loop-unrolled SAR ADC with common-mode adaptive background offset calibration in 28 nm FDSOI," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 7, pp. 2766–2774, Jul. 2021.
[5] Y. Liang, R. Ding, and Z. Zhu, "A 9.1 ENOB 200 MS/s asynchronous SAR ADC with hybrid single-ended/differential DAC in 55-nm CMOS for image sensing signals," *IEEE Sensors J.*, vol. 18, no. 17, pp. 7130–7140, Sep. 2018.
[6] M. Li *et al.*, "A 6.94-fJ/conversion-step 12-bit 100-MS/s asynchronous SAR ADC exploiting split-CDAC in 65-nm CMOS," *IEEE Access*, vol. 9, pp. 77545–77554, 2021.
[7] S. Sunny, Y. Chen, and C. C. Boon, "A 4.06 mW 10-bit 150 MS/s SAR ADC with 1.5-bit/cycle operation for medical imaging applications," *IEEE Sensors J.*, vol. 18, no. 11, pp. 4553–4560, Jun. 2018.
[8] M. Zhang, C. H. Chan, Y. Zhu, and R. P. Martins, "A 0.6-V 13-bit 20-MS/s two-step TDC-assisted SAR ADC with PVT tracking and speed enhanced techniques," *IEEE J. Solid-State Circuits*, vol. 54, no. 12, pp. 3396–3409, Dec. 2019.
[9] Z. Yu, Y. Liang, and S. Liu, "A 5 MHz-BW 71.7-dB SNDR two-step hybrid-domain ADC in 65-nm CMOS," *Microelectron. J.*, vol. 117, Nov. 2021, Art. no. 105253.
[10] M. Ni, X. Wang, F. Li, and Z. Wang, "A 13-bit 312.5-MS/s pipelined SAR ADC with open-loop integrator-based residue amplifier and gain stabilized integration time generation," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 29, no. 7, pp. 1416–1427, Jul. 2021.
[11] P. Yi, Y. Liang, S. Liu, N. Xu, L. Fang, and Y. Hao, "A 625kHz-BW, 79.3dB-SNDR second-order noise-shaping SAR ADC using high-efficiency error-feedback structure," *IEEE Trans. Circuits Syst. II, Exp. Briefs.*, vol. 69, no. 3, pp. 859–863, Mar. 2022.
[12] C. Yang *et al.*, "An area-efficient SAR ADC with mismatch error shaping technique achieving 102-dB SFDR 90.2-dB SNDR over 20-kHz bandwidth," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 29, no. 8, pp. 1575–1585, Aug. 2021.
[13] M. Ding, P. Harpe, Y. H. Liu, B. Busze, K. Philips, and H. de Groot, "A 46 μW 13 b 6.4 MS/s SAR ADC with background mismatch and offset calibration," *IEEE J. Solid-State Circuits*, vol. 52, no. 2, pp. 423–432, Feb. 2017.
[14] K. H. Chang and C. C. Hsieh, "A calibration-free 12-bit 50-MS/s full analog SAR ADC with feedback zero-crossing detectors," *IEEE J. Solid State Circuits*, vol. 54, no. 6, pp. 1624–1635, Jun. 2019.
[15] M. Ding *et al.*, "A hybrid design automation tool for SAR ADCs in IoT," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 26, no. 12, pp. 2853–2862, Dec. 2018.
[16] C. Wulff and T. Ytterdal, "A compiled 9-bit 20-MS/s 3.5-fJ/conv.step SAR ADC in 28-nm FDSOI for Bluetooth low energy receivers," *IEEE J. Solid-State Circuits*, vol. 52, no. 7, pp. 1915–1926, Jul. 2017.
[17] J. Park, Y. Hwang, and D. Jeong, "A 0.5-V fully synthesizable SAR ADC for on-chip distributed waveform monitors," *IEEE Access*, vol. 7, pp. 63686–63697, 2019.
[18] M.-J. Seo, Y.-J. Roh, D.-J. Chang, W. Kim, Y.-D. Kim, and S.-T. Ryu, "A reusable code-based SAR ADC design with CDAC compiler and synthesizable analog building blocks," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 65, no. 12, pp. 1904–1908, Dec. 2018.
[19] Z. Xu, N. Ojima, S. Li, and T. Iizuka, "An all-standard-cell-based synthesizable SAR ADC with nonlinearity-compensated RDAC," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 29, no. 12, pp. 2153–2162, Dec. 2021.
[20] S. Li, B. Xu, D. Z. Pan, and N. Sun, "A 60-fJ/step 11-ENOB VCO based CTDSM synthesized from digital standard cell library," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, Apr. 2019, pp. 1–4.
[21] P. H. Wei and B. Murmann, "Analog and mixed-signal layout automation using digital place-and-route tools," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 29, no. 11, pp. 1838–1849, Nov. 2021.
[22] P. Harpe *et al.*, "A 26 μW 8 bit 10 MS/s asynchronous SAR ADC for low energy radios," *IEEE J. Solid-State Circuits*, vol. 46, no. 7, pp. 1585–1595, Jul. 2011.
[23] P. Harpe, "A compact 10-b SAR ADC with unit-length capacitors and a passive FIR filter," *IEEE J. Solid-State Circuits*, vol. 54, no. 3, pp. 636–645, Mar. 2019.
[24] C. C. Lee, C. Y. Lu, R. Narayanaswamy, and J. B. Rizk, "A 12b 70MS/s SAR ADC with digital startup calibration in 14nm CMOS," in *Proc. Symp. VLSI Circuits*, Jun. 2015, pp. C62–C63.
[25] J. Shen *et al.*, "A 16-bit 16-MS/s SAR ADC with on-chip calibration in 55-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 53, no. 4, pp. 1149–1160, Apr. 2018.
[26] A. M. Abo and P. R. Gary, "A 1.5-V, 10-bit, 14.3-MS/s CMOS pipeline analog-to-digital converter," *IEEE J. Solid-State Circuits*, vol. 34, no. 5, pp. 599–606, May 1999.
[27] B. Razavi, "The design of a bootstrapped sampling circuit," *IEEE Solid-State Circuits Mag.*, vol. 13, no. 1, pp. 7–12, Winter 2021.
[28] *Cadence Abstract Generator User Guide*, Cadence Des. Syst., San Jose, CA, USA, Jan. 2004.
[29] B. Razavi, "The StrongARM latch," *IEEE Solid-State Circuits Mag.*, vol. 7, no. 2, pp. 12–17, Jun. 2015.
[30] *Calibre Verification User's Manual*, Cadence Des. Syst., San Jose, CA, USA, May 2002.
[31] *Encounter User Guide*, Cadence Des. Syst., San Jose, CA, USA, May 2005.
[32] *Virtuoso NC-Verilog Environment User Guide*, Cadence Des. Syst., San Jose, CA, USA, Aug. 2021.
[33] Y. Shen *et al.*, "A 10-bit 120-MS/s SAR ADC with reference ripple cancellation technique," *IEEE J. Solid-State Circuits*, vol. 55, no. 3, pp. 680–692, Mar. 2020.



Yao-Hung Tsai was born in Taichung, Taiwan, in 1998. He received the B.S. degree in electrical and computer engineering from National Chiao Tung University, Hsinchu, Taiwan, in 2020. He is currently working toward the Ph.D. degree in electronics engineering at National Taiwan University, Taipei, Taiwan.

His research interests include high-resolution data converters and mixed-signal integrated circuits.



Shen-Iuan Liu (Fellow, IEEE) was born in Keelung, Taiwan, in 1965. He received the B.S. and Ph.D. degrees in electrical engineering from National Taiwan University (NTU), Taipei, Taiwan, in 1987 and 1991, respectively.

From 1991 to 1993, he has served as a Second Lieutenant for the Chinese Air Force, Tatung, Taiwan. From 1991 to 1994, he was an Associate Professor with the Department of Electronic Engineering, National Taiwan Institute of Technology, Taipei. He joined the Department of Electrical Engineering, NTU, in 1994, where he has been a Professor since 1998. He has been a Distinguished Professor with NTU since August 2010. He was the Director of the Graduate Institute of Electronics Engineering, NTU, from 2013 to 2016. His research interests include analog and digital integrated circuits and systems.

Dr. Liu has served as a Technical Program Committee Member for International Solid-State Circuits Conference (ISSCC) from 2006 to 2008, the IEEE International Symposium on VLSI Design, Automation and Test (VLSI-DAT) from 2008 to 2012, and Asia Solid-State Circuits Conference (A-SSCC) from 2005 to 2012. He was a recipient of the Engineering Paper Award from the Chinese Institute of Engineers in 2003, the Young Professor Teaching Award from MXIC Inc., the Research Achievement Award from NTU, the Outstanding Research Award from the National Science Council in 2004, and the Outstanding Research Award from the Ministry of Science and Technology in 2014. He was awarded the Himax Chair Professorship at NTU in 2010. He was also a recipient of the Best Paper Award at the 2020 and 2021 International Symposium on VLSI Design, Automation and Test. He has also served as the Chair for the IEEE SSCS Taipei Chapter from 2004 to 2008, which achieved the Best Chapter Award in 2009. He has also served as the General Chair for the 15th VLSI Design/CAD Symposium, Taiwan, in 2004, and the Program Co-Chair for the Fourth IEEE Asia-Pacific Conference on Advanced System Integrated Circuits, Fukuoka, Japan, in 2004. He has also served as the Technical Program Committee Co-Chair and the Chair for A-SSCC 2010 and 2011, respectively. He was an Associate Editor of IEEE JOURNAL OF SOLID-STATE CIRCUITS from 2006 to 2009 and the Guest Editor of IEEE JOURNAL OF SOLID-STATE CIRCUITS Special Issue in December 2008 and November 2012. He was an Associate Editor of IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS from 2006 to 2007. He was an Associate Editor of IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—I: REGULAR PAPERS from 2008 to 2009. He was the Editorial Board of *Research Letters in Electronics* from 2008 to 2009. He was an Associate Editor of The Institute of Electronics, Information and Communication Engineers (IEICE) Transactions on Electronics from 2008 to 2011. He has been an Associate Editor of *ETRI Journal*. He has also been an Associate Editor of *Journal of Semiconductor Technology and Science*, South Korea, since 2009.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on February 04,2026 at 06:35:05 UTC from IEEE Xplore. Restrictions apply.